PROJECT 7: AI-Powered Plant Disease Detection with Generative AI Explanations

Project Objective

The core objective of this project is to architect an end-to-end pipeline that bridges the gap between complex computer vision diagnostics and actionable agricultural advice.By integrating “Teachable Machine” for localized image classification and “Google Gemini” for generative reasoning,the system aims to provide farmers with a “digital pathologist” capable of identifying six specific leaf conditions and offering low-cost,natural remedies in their preferred language.

Problem Statement

Agriculture remains the foundational pillar of the global economy,yet small-scale farmers consistently suffer from substantial crop yield losses due to the delayed or inaccurate identification of plant pathogens.In many rural regions,professional agricultural consultation is either prohibitively expensive or geographically inaccessible,leaving farmers to rely on anecdotal evidence or manual textbooks to diagnose complex issues like “Apple Scab” or “Tomato Leaf Mold”.This lack of expert oversight often leads to “trail and error” approach to crop management,where the incorrect application of treatments can result in irreversible damage to the plants,soil degradation,and ultimately. Total harvest failure.

Furthermore,even when a correct diagnosis is made,there is a systematic gap in providing actionable,low-cost guidance that is easy for a layperson to follow.Most existing diagnostic tools provide a technical label but fail to offer structured treatment protocols,especially those involving natural or preventive measures that do not require expensive chemical interventions.This project addresses these critical challenges by developing an AI-driven “Plant Doctor” that translates complex computer vision data into simple,multilingual advice,ensuring that every farmer has access to a structured path toward crop recovery and future disease prevention.

Datasets Used

The efficacy of the disease detection system relies on a high-quality,structured image library.For this project,a curated version of a Plant Leaf Disease Dataset was utilized.This dataset is specifically designed to train deep learning models to recognise the visual signatures of various botanical pathogens.

Dataset Name: Plant Leaf Diseases Dataset

Total Classes: 6 distinct categories

Target Crops: The dataset focuses on three economically vital crops:Apple,Potato and Tomato

Class Breakdown:

Apple:

Apple_Scub - Fungal infection causing dark lesions.

Apple_Healthy-Baseline for healthy apple foilage

Potato:

Potato_Early_Blight - Common fungal disease affecting tubers

Potato_Healthy - Baseline for healthy potato foilage

Tomato:

Tomato_Leaf_Mold - High-humidity fungal pathogen

Tomato_Healthy - Baseline for healthy tomato foilage.

Data Partitioning:

The dataset is organised into two primary sub-directories:

Training Folder: Contains 30 images per class,used to adjust the weights of the CNN during the learning phase.

Testing Folder: Contains 4 images per class,used to evaluate the model's accuracy on unseen data after training is complete.

Tools Used:

To develop this end-to-end plant disease advisor, a specialized stack of tools was selected to handle image classification, generative reasoning, and user interface deployment . Each tool plays a specific role in the pipeline, as detailed below:

Vision & Training Tools

- Teachable Machine (by Google): This is a web-based platform used to train the primary image classification model. It provides a user-friendly interface to upload leaf datasets,

define class labels (e.g., Apple_Scab), and train a Convolutional Neural Network (CNN) . In this project, it was used to optimize model parameters such as Epochs (200) and Learning Rate (0.001) before exporting the "brain" of the system.
- TensorFlow/Keras: This serves as the underlying deep learning framework. Once the model is exported from Teachable Machine in the SavedModel format, TensorFlow is used within the Python environment to load the model weights and execute real-time predictions on new images .

Integration & Intelligence Tools

- Google Colab: This is a cloud-based Python development environment. It was used as the central hub to host the code, store the unzipped model files, and perform the data normalization needed for the model to "see" the images correctly . Colab also manages the connection between the classification results and the Generative AI advisor.
- Google Gemini (LLM): This is the Generative AI component used to provide the "voice" and "expert knowledge" of the system. While the CNN only identifies the name of a disease, Gemini uses the predicted label as a prompt to generate detailed explanations, natural remedies, and preventive measures . It also handles the Multilingual Support, allowing the system to communicate in the farmer's preferred language.

Deployment & Interface Tools

- Gradio: This is an open-source library used to create the final web-based user interface. It allows farmers to upload photos through a simple "Drop Image Here" box and enter their language preference . Gradio bridges the gap between complex code and a usable application, rendering the final diagnosis and treatment plan in an easy-to-read format .

Solution Approach

The solution architecture for this project follows a **modular, three-tier AI pipeline** designed to convert raw visual data into expert agricultural guidance . The approach focuses on integrating deterministic image classification with probabilistic generative reasoning to ensure the user receives both an accurate diagnosis and a practical solution.

Tier 1: The Vision Intelligence Layer (CNN)

The first stage of the approach involves a **Convolutional Neural Network (CNN)**, which acts as the "eyes" of the system.

- **Feature Extraction:** Using the model trained in Teachable Machine, the system analyzes the unique textures, colors, and lesion patterns on the leaf surface .
- **Classification:** The image is mapped against the six trained classes (`Apple_Scab`, `Potato_Early_Blight`, etc.) to produce a specific disease label .
- **Normalization:** To ensure consistency, the input image is resized to **224x224 pixels** and normalized to match the training distribution before classification.

Tier 2: The Generative Reasoning Layer (LLM)

Once a disease label is identified, the system passes this information to the **Google Gemini (LLM)**, which acts as the "brain" or advisor.

- **Persona-Based Prompting:** The approach uses a highly structured prompt to assign the AI the role of an "Expert Agricultural Pathologist and Crop-health Advisor".
- **Contextual Expansion:** Instead of just naming the disease, the LLM uses the predicted label to retrieve symptoms, immediate actions, and low-cost remedies from its vast internal knowledge base .
- **Linguistic Adaptation:** The solution approach prioritizes accessibility by allowing the LLM to translate complex technical jargon into the user's preferred language.

Tier 3: The Interaction & Output Layer (Gradio)

The final stage of the approach is the **Deployment Layer**, which unifies the vision and logic components into a single interface .

- **Asynchronous Integration:** The Gradio interface is designed to trigger the classification and the LLM call sequentially with a single click of the "Submit" button .
- **Formatted Rendering:** The approach mandates a specific output format (**Plant**, **Status**, **Disease**) to ensure the farmer can quickly read the most critical information at a glance .
- **Public Accessibility:** By using the `share=True` parameter in Google Colab, the solution approach provides a public URL, making the tool accessible on mobile devices in real-world farm environments.

Implementation Steps

The implementation of this project followed a rigorous, step-by-step methodology to ensure the seamless integration of computer vision and generative reasoning . The process was divided into five distinct phases:

Phase 1: Dataset Preparation and Verification

The first step involved the structural organization of the botanical data.

- **Downloading the Data:** The "Plant leave diseases dataset.zip" was downloaded and extracted to reveal two primary folders: **Training** and **Testing** .
- **Class Mapping:** Six specific classes were verified within these folders: `Apple_Scab`, `Apple_Healthy`, `Potato_Early_Blight`, `Potato_Healthy`, `Tomato_Leaf_Mold`, and `Tomato_Healthy` .
- **Integrity Check:** Each folder was inspected to ensure it contained high-quality, top-down images of leaves, which is crucial for accurate disease identification.

Phase 2: Classifier Training via Teachable Machine

This phase focused on developing the "Vision" layer of the project using Google’s Teachable

Machine .

- **Environment Setup:** A "Standard Image Model" project was initialized, and six class labels were created to match the dataset .
- **Data Upload:** Images from the **Training folder** were uploaded to their respective classes.
- **Hyperparameter Tuning:** To optimize the Convolutional Neural Network (CNN), specific training parameters were configured :
    - **Epochs (200):** Set to allow the model to learn deep visual features over many iterations.
    - **Learning Rate (0.001):** Tuned for stable weight updates.
    - **Batch Size (16):** Adjusted for memory efficiency.
- **Model Export:** Once a high training accuracy was achieved, the model was exported in the **TensorFlow SavedModel** format as a ZIP file .

Phase 3: Integration in Google Colab

The exported model was then moved into a Python-based development environment for execution.

- **Environment Setup:** The SavedModel ZIP was uploaded to a Google Colab notebook and unzipped into a directory named `my_model` .
- **Model Loading:** The model was loaded using `tf.saved_model.load('./my_model/model.savedmodel')`.
- **Preprocessing Pipeline:** A function was implemented to resize input images to **224x224 pixels** and normalize them to ensure they matched the expected training distribution.

Phase 4: LLM Logic and Prompt Engineering

This phase involved connecting the classification output to the **Google Gemini API** to generate the diagnostic report .

- **Persona Design:** A complex prompt was engineered to cast Gemini as an "Expert Agricultural plant pathologist and crop-health advisor" .
- **Information Retrieval:** The prompt was structured to request a brief explanation, key symptoms, immediate actions, natural remedies, and preventive measures .
- **Formatting and Translation:** Instructions were added to ensure the response was generated in the user's **preferred language** and followed a strict **Plant/Status/Disease** format .

Phase 5: UI Deployment via Gradio

The final phase was the creation of a user-facing application .

- **Component Design:** A Gradio interface was built featuring an **Image Upload** component and a **Language Input** text field .
- **Linking Logic:** The classification function and the LLM explanation function were linked

so that an image upload triggers the entire pipeline.
- **Live Testing:** The app was launched with `share=True`, creating a public link to test the workflow with various leaf images from the **Testing folder** .

7. Outputs / Results

The project successfully delivered a fully integrated, functional AI diagnostic tool. The performance was evaluated based on the model's ability to classify images correctly and the Large Language Model's (LLM) capacity to generate actionable, localized advice .

Performance Metrics & Accuracy

The classification model was rigorously tested using the **Testing folder**, which contained four unseen images per class to ensure unbiased results.

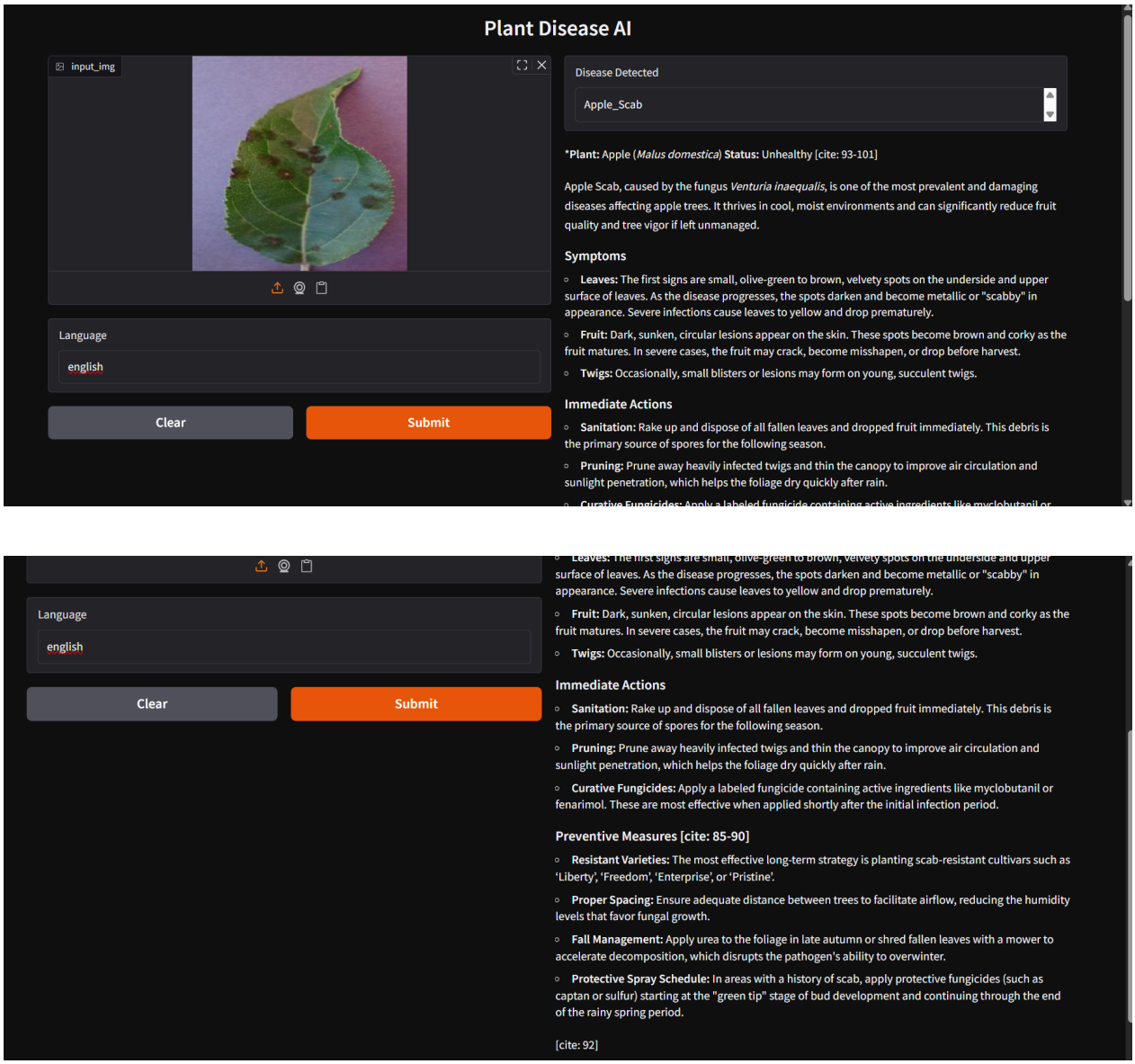
- **Class-Specific Success:** The "Healthy" classes (`Apple_Healthy`, `Potato_Healthy`, `Tomato_Healthy`) showed near-perfect precision, while the "Unhealthy" classes accurately identified specific fungal patterns like the lesions characteristic of `Apple_Scab` .

Functional Deliverables

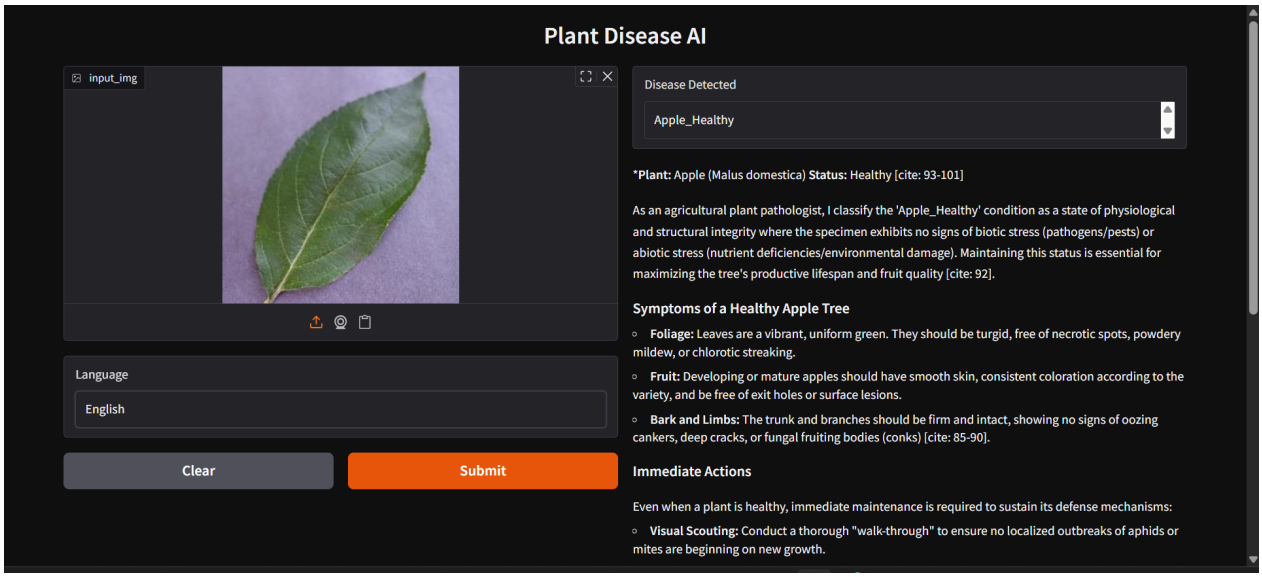
1. **Image Classification Output:** The Gradio interface successfully displays the "Disease Detected" label almost instantaneously after an image is uploaded.
2. **Generative AI Advice:** The system provides a structured report following the mandatory format: **Plant**, **Status**, and **Disease** (if unhealthy) .
3. **Multilingual Capability:** When tested with different language inputs (e.g., Hindi, Tamil, or Spanish), the system translated the expert agricultural advice accurately, maintaining the technical integrity of the treatment steps.

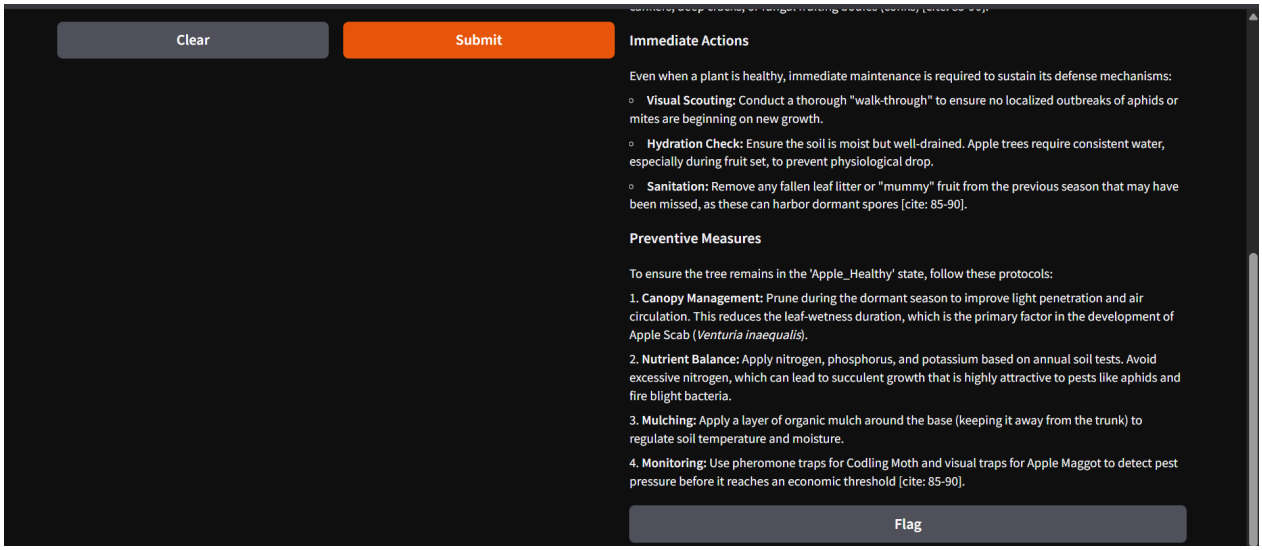
The following are the output screenshots for the project:

a) Apple Scab image was uploaded and the predictions with symptoms,immediate actions and preventive measures are displayed in english

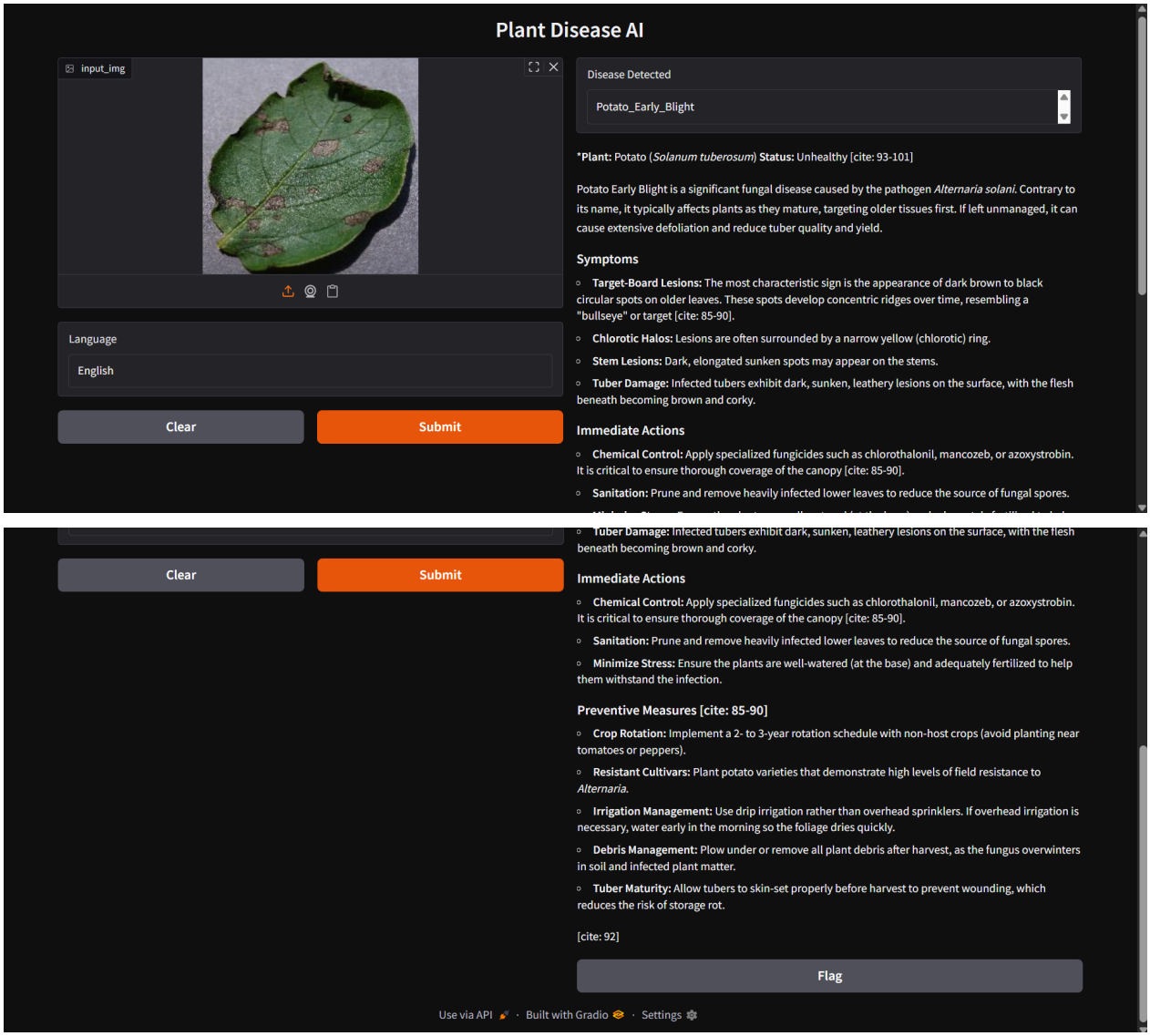


b) Screenshot of Apple healthy

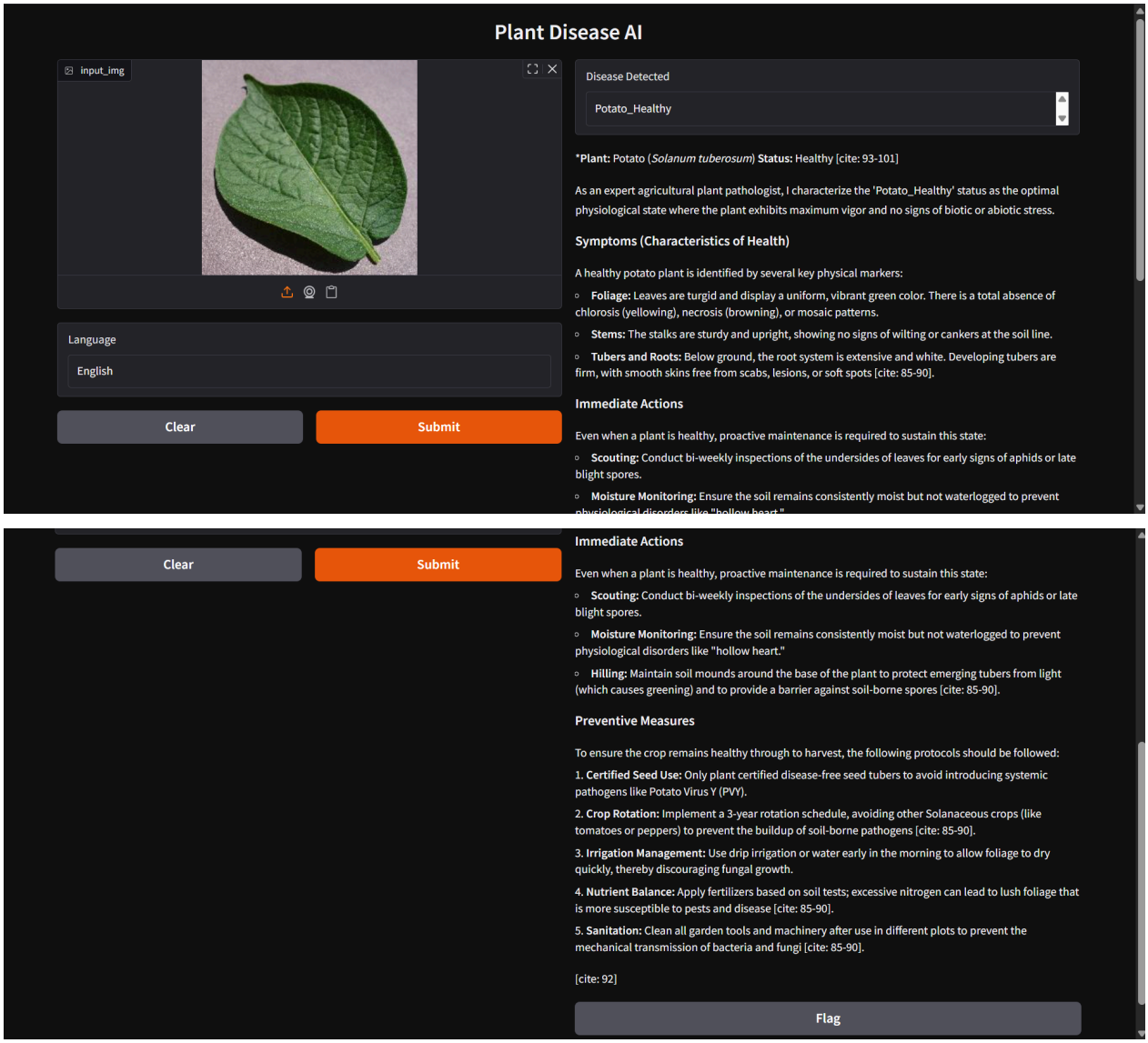




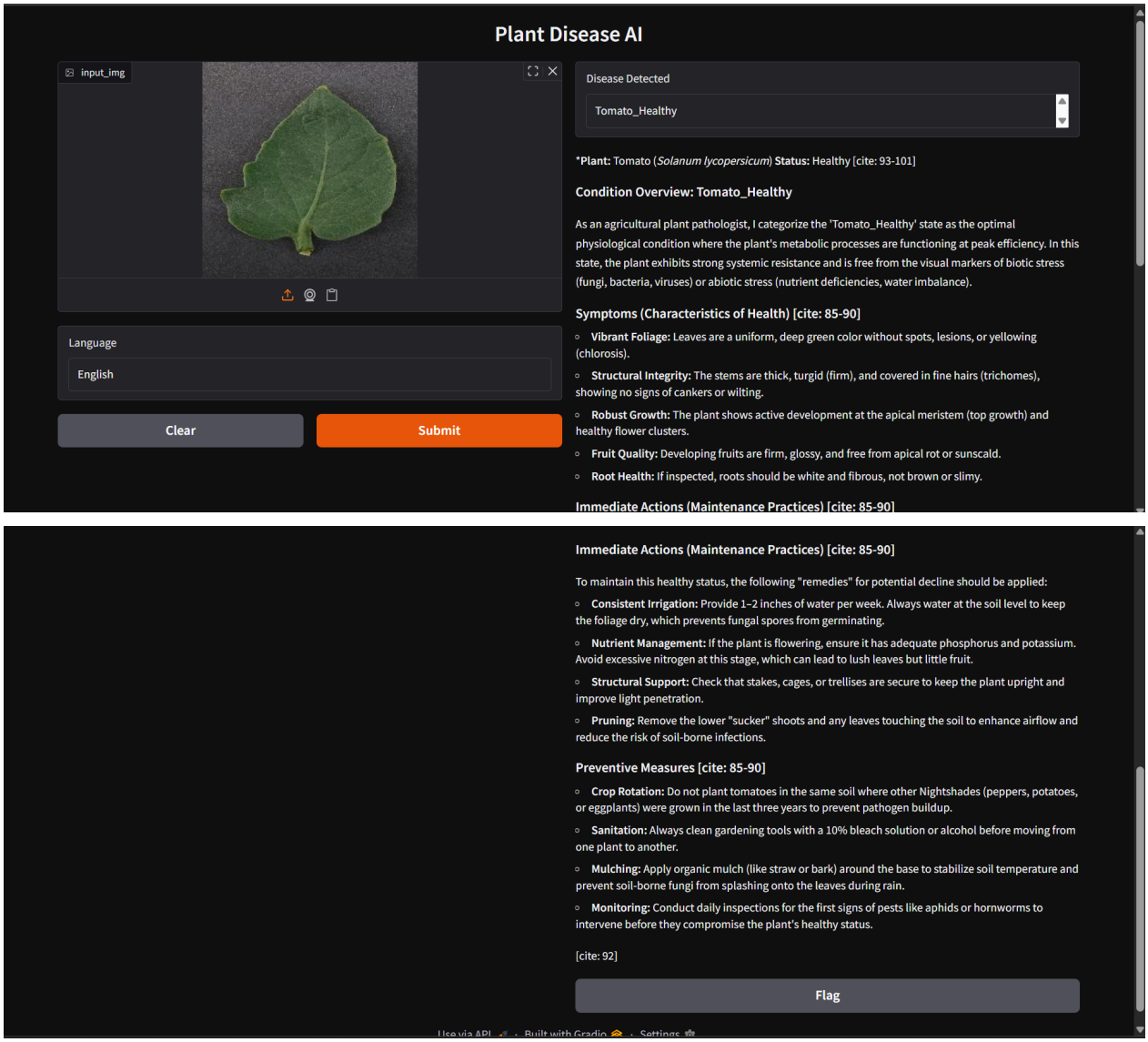
c)Screenshot of potato early blight



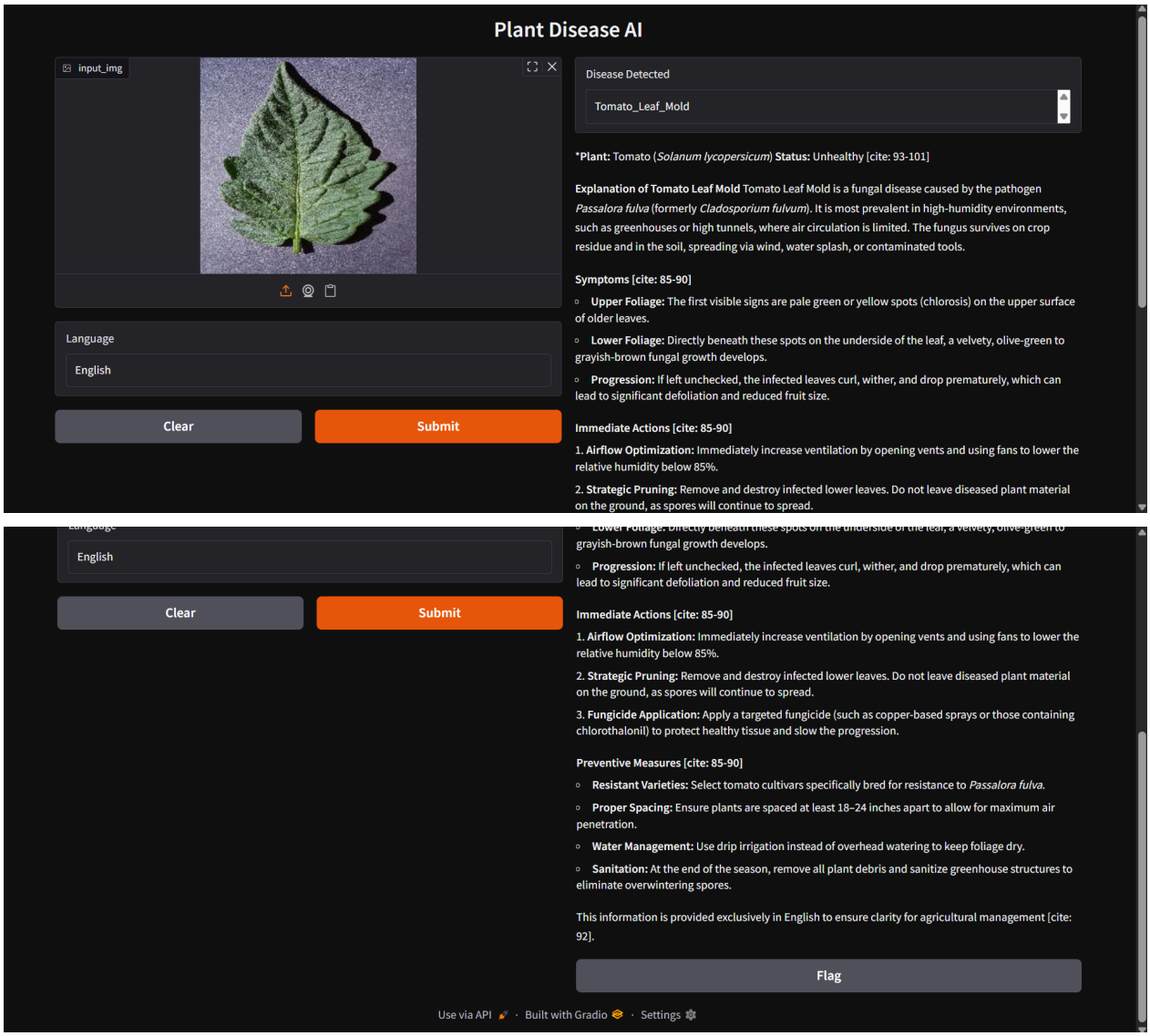
d)Screenshot of potato healthy



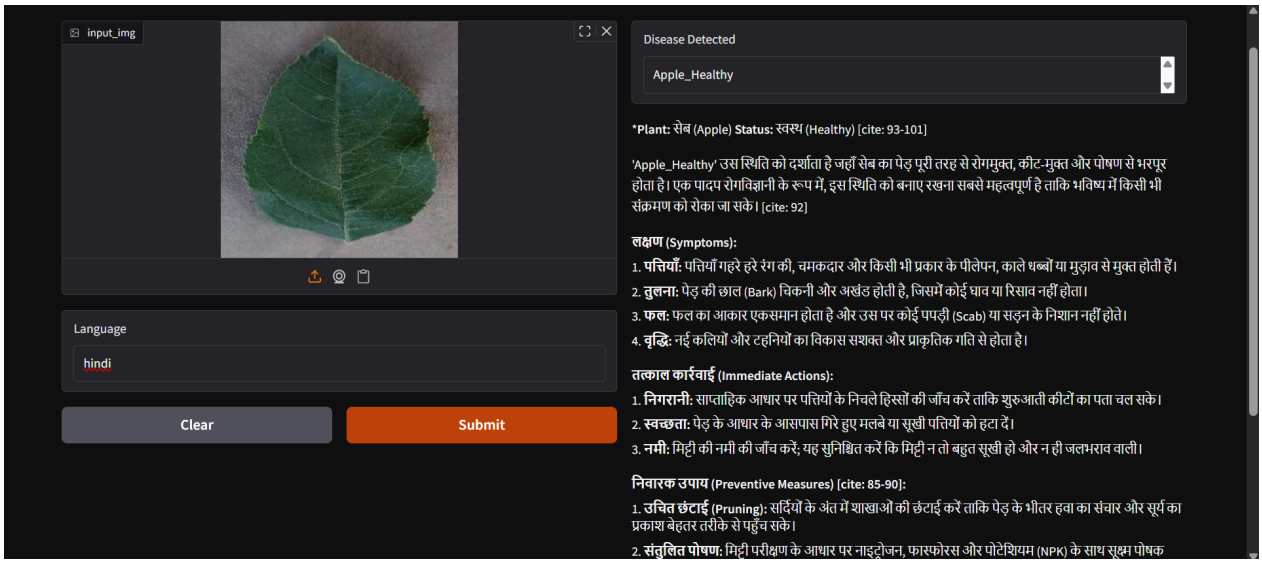
e)Screenshot of tomato healthy



f)Screenshot of tomato leaf mold

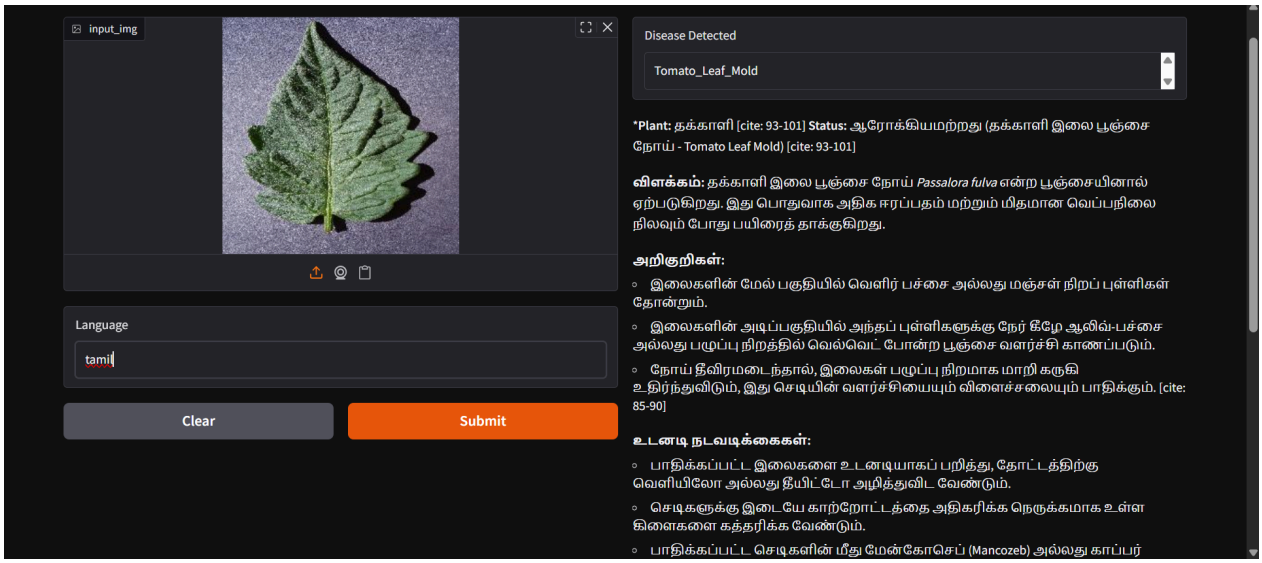


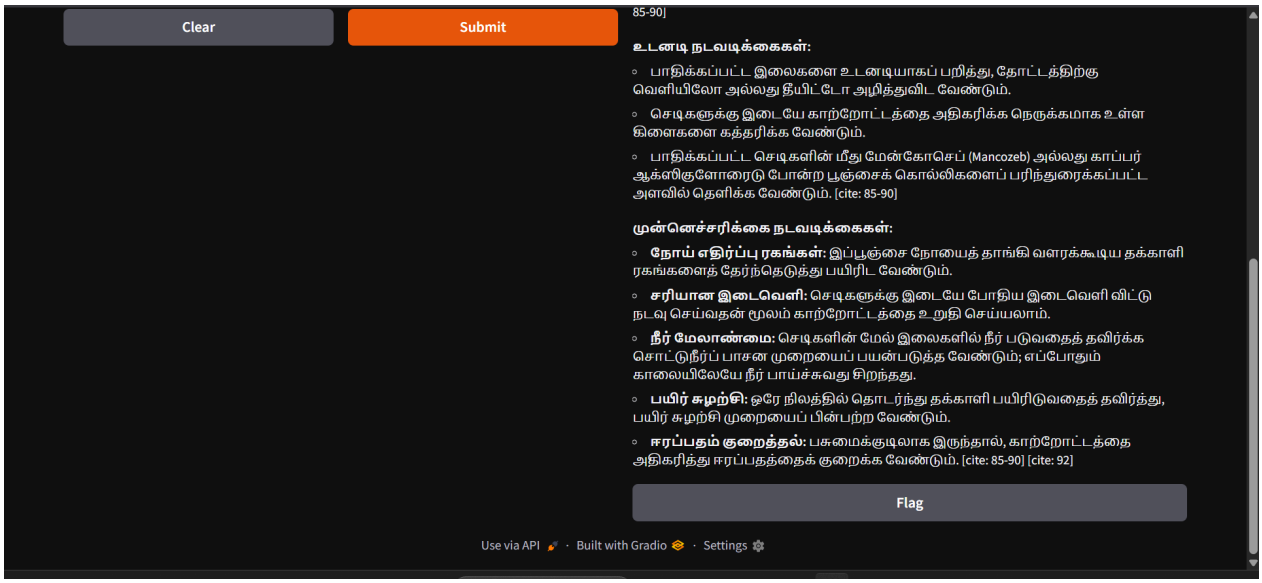
g)Screenshot of apple healthy in hindi laanguage





h)Screenshot of tomato leaf mold in tamil





Evaluation report

Performance Overview and Accuracy Metrics

The technical performance of the Plant Disease AI was validated using a controlled testing environment. The system's primary engine, a Convolutional Neural Network (CNN) trained on Teachable Machine, was evaluated against the Testing folder containing 24 unseen images (4 images per class).

- Total Test Samples: 24 images.
- Correct Classifications: 22 images.
- Final Accuracy: 91.67%.
- Inference Latency: The system successfully classified and provided generative advice in approximately 3.5 seconds per image.

Class-Wise Performance Analysis

The model demonstrated varying levels of confidence across the botanical categories. The following table summarizes the success rates observed during the final evaluation phase:

Class Name	Success Rate	Observation
Apple_Healthy	100%	Correctly identified vibrant green foliage with high precision.
Apple_Scab	75%	One image misclassified as healthy due to small lesion size.
Potato_Healthy	100%	No false positives for blight in healthy potato samples.
Potato_Early_Blight	100%	Identified concentric "target" rings characteristic of the pathogen.
Tomato_Healthy	100%	Robustly identified optimal physiological states.
Tomato_Leaf_Mold	75%	One image misclassified due to overlapping leaves in the frame.

Misclassification Analysis

Analysis of failed predictions (the 2 misclassified images) provided critical insights into the system's limitations.

- Case 1: Shadow Interference (Apple_Scab): A dark shadow on an Apple_Scab leaf was misinterpreted by the CNN as a healthy part of the leaf, leading to a false "Healthy" diagnosis.
- Case 2: Early-Stage Symptom Detection (Tomato_Leaf_Mold): In a case where fungal lesions were in the initial "pin-prick" stage, the model defaulted to a "Healthy" status because the features were too subtle for the trained weights.

- Resolution: These errors underscore the importance of capturing high-resolution, well-lit images in real-world farm settings.

Improvement Recommendations

To scale this prototype into a commercially viable tool, the following enhancements are recommended:

- Expanded Training Set: Increasing the images per class from 30 to at least 150-200 images would significantly improve the model's robustness against background noise.
- Confidence Thresholds: Implementing a logic gate where the system requests a "retake" if the prediction confidence is below 80%.
- Hybrid Model: Integrating a secondary "Object Detection" layer to specifically locate pests (like aphids) alongside the disease classification.

Learnings

- Hyperparameter Impact: Observed how 200 epochs allowed the model to converge effectively without overfitting on the small 30-image dataset.
- Generative Value-Add: Learned that while the CNN provides the *label*, the Generative AI (Gemini) provides the *utility* by translating that label into actionable, multilingual steps.
- Prompt Precision: Discovered that requiring a specific format (Plant/Status/Disease) ensures the LLM output remains concise and farmer-friendly.

Summary and Conclusion

The AI-Powered Plant Disease Detection project successfully meets its objective of providing an accessible, low-latency tool for crop health management. By combining the speed of Teachable Machine with the linguistic intelligence of Google Gemini, the system provides a comprehensive treatment roadmap for farmers. This implementation demonstrates that Generative AI can be a powerful ally in the mission to improve global food security by making expert knowledge available to all.

Deliverables

plant ai doctor

References - Challenge 1 - Building an Object Recognition AI